Let, $\lfloor x \rfloor = m$ and $\lceil x \rceil = n$. By Defn. 12, m and n are integers.

$\therefore x = m + \varepsilon_1$, where $0 \leq \varepsilon_1 < 1$ and $x = n - \varepsilon_2$, where $0 \leq \varepsilon_2 < 1$.

$x - x = (n - \varepsilon_2) - (m + \varepsilon_1) \Rightarrow 0 = (n - m) - (\varepsilon_2 + \varepsilon_1) \Rightarrow n - m = \varepsilon_2 + \varepsilon_1$

We know, $n - m \in Z^+$ $\because$ by Prop (2), $\lfloor x \rfloor < \lceil x \rceil$ $\quad \varepsilon_1 + \varepsilon_2 > 0$ and $\varepsilon_1 + \varepsilon_2 < 2$.

$\therefore \varepsilon_1 + \varepsilon_2 = 0$ or 1. If $\varepsilon_1 + \varepsilon_2 = 0$, then $\varepsilon_1 = \varepsilon_2 = 0$.

$\therefore x = n - 0 \Rightarrow x = n$ $\quad x = m + 0 \Rightarrow x = m$. $\therefore m = n \Rightarrow n - m = 0$

$\Rightarrow \boxed{\lceil x \rceil - \lfloor x \rfloor = 0}$

In this case, x is an integer.

If $\varepsilon_1 + \varepsilon_2 = 1$. Suppose x is not an integer. $\therefore \varepsilon_1$ and $\varepsilon_2 > 0$.

$\varepsilon_1 + \varepsilon_2 > 0 + 0 = 0$. $\therefore \varepsilon_1 + \varepsilon_2 = 1$. $\therefore n - m = 1 \Rightarrow \boxed{\lceil x \rceil - \lfloor x \rfloor = 1}$ □

51) Show that if x is a real number, then $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$.

Soln: Let, $\lfloor x \rfloor = m$ and $\lceil x \rceil = n$. By Defn. 12, m and n are integers.

By Prop (1c), $x - 1 < m \leq x$. By Prop 1(d), $x \leq n < x + 1$.

$\therefore x - 1 < m \leq x \leq n < x + 1 \Rightarrow \boxed{x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1}$ □

52) Show that if x is a real number and m is an integer, then $\lceil x + m \rceil = \lceil x \rceil + m$

Soln: Let, $\lceil x + m \rceil = n$. By Defn. 12, n is an integer.

By Prop (1b), $n - 1 < x + m \leq n$. $\Rightarrow (n - m) - 1 < x \leq n - m$

$\therefore$ By Prob (1b), $\lceil x \rceil = n - m \Rightarrow n = \lceil x \rceil + m \Rightarrow \lceil x + m \rceil = \lceil x \rceil + m$.

53) Show that if x is a real number and n is an integer, then

a) $x < n$ if and only if $\lfloor x \rfloor < n$.

Soln: Let, $x < n$. By Property 2 from Table 1, $\lfloor x \rfloor \leq x$.

$\therefore \lfloor x \rfloor < n$.

Let, $\lfloor x \rfloor < n$. Let, $\lfloor x \rfloor = m$. By Defn. 12, m is an integer.

By Property (1a) from Table 1, $m \leq x < m + 1$.

Given, x is a real no. $\therefore m < x < m + 1$. $\because m + 1 \in Z$, $\therefore n \geq m + 1$.

$\therefore \boxed{x < n}$ □